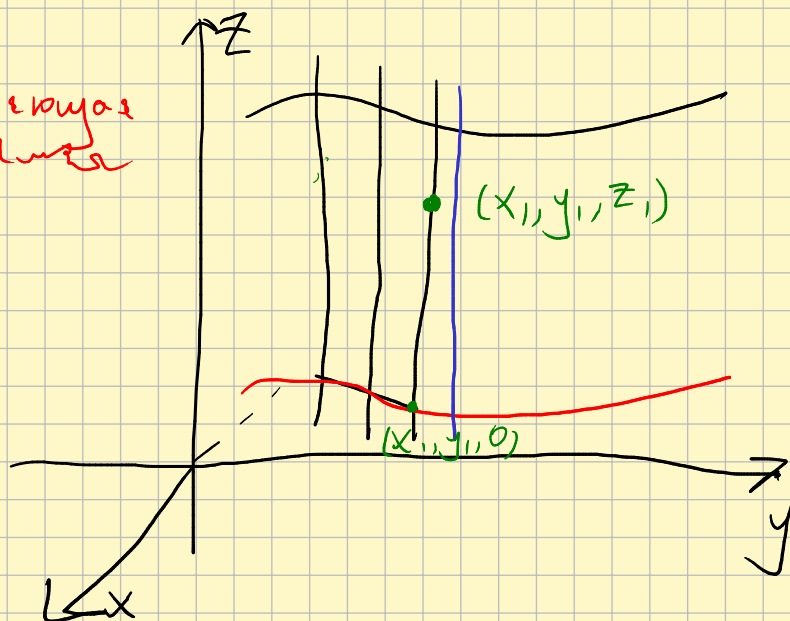


Цилиндрическая поверхность

$f(x, y) = 0$ направляющая линия

— образ-л при.

$$F(x, y, z) = 0$$



$$1. \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$2. \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

$$3. \frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

$$4. \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0$$

$$x = h$$

$$\frac{h^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$

$$\frac{h^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}, \quad \frac{y^2}{b^2} - \frac{z^2}{c^2} = -\frac{h^2}{a^2}$$

$$5. \frac{x^2}{a^2} + \frac{y^2}{b^2} = z$$

$$6. \frac{x^2}{a^2} - \frac{y^2}{b^2} = z$$

$$7. \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$8. \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$9. \frac{x^2}{a^2} + \frac{y^2}{b^2} = 0 \leftarrow \text{прямая}$$

$$9.7. \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0 \leftarrow 2 \text{ плоскости}$$

$$10. y^2 = 2bx \leftarrow \text{парабола}$$

$$11. x^2 = 1 \leftarrow \pi_1, \pi_2$$

$$12. x^2 = 0 \leftarrow \pi_1 = \pi_2$$

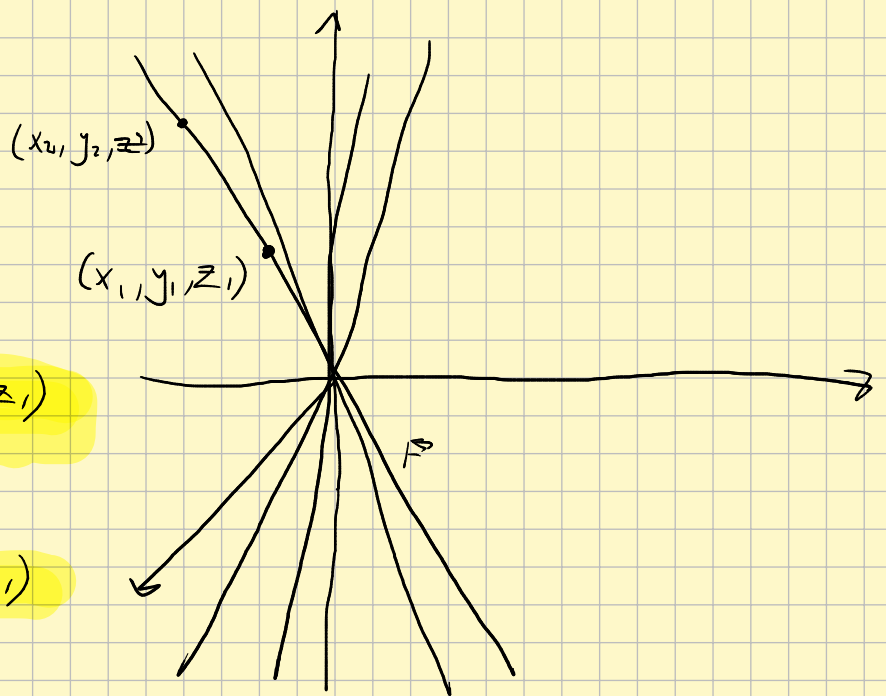
$$\begin{cases} x_2 = \lambda x_1, \\ y_2 = \lambda y_1, \\ z_2 = \lambda z_1. \end{cases}$$

$$F(\lambda x_1, \lambda y_1, \lambda z_1) = \lambda^2 F(x_1, y_1, z_1)$$

$$F(\lambda x, \lambda y, \lambda z) = \lambda^n F(x, y, z)$$

$\Downarrow$

$$(\lambda x, \lambda y, \lambda z) \in F$$



$$\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1 - \frac{y^2}{b^2} \Rightarrow \left(\frac{x}{a} - \frac{z}{c} \right) \left(\frac{x}{a} + \frac{z}{c} \right) = \left(1 - \frac{y}{b} \right) \left(1 + \frac{y}{b} \right)$$

$\Downarrow$

$$\begin{cases} \frac{x}{a} - \frac{z}{c} = 1 - \frac{y}{b}, \\ \frac{x}{a} + \frac{z}{c} = 1 + \frac{y}{b} \end{cases}$$

